DAILY PROGRESS REPORT - DAY 1 {16.04.26}
Autonomous Navigation and Load Balancing System for Military Quadruped Robots

1. Report Information

NAME : HARINI P (DEPARTMENT OF COMPUTER SCIENCE AND ENGNEERING)

TARGET JOURNAL : IEEE Robotics & Automation Letters (IEEE RA-L)

2. Nature of Work Done Today

Today marked the official start (Day 1) of the 29-day journal paper conversion project. The primary focus was **foundational planning and research** — selecting the appropriate target journal, understanding its submission requirements, studying comparable published papers, and preparing the working draft document. No coding or simulation work was performed today; this was a structured research and planning session essential before writing begins.

3. Tasks Completed

Task / Activity	Description	Status
Journal Selection	Searched IEEE Xplore & Google Scholar; selected IEEE Robotics & Automation Letters (IEEE RA-L) as the target journal based on scope, impact factor, and similar published work on quadruped robotics.	✓ **Completed**
Author Guidelines Review	Downloaded and carefully reviewed the IEEE RA-L Author Guidelines PDF including page limits (up to 8 pages), citation style (IEEE), figure count requirements, and submission checklist.	✓ **Completed**
Sample Paper Study	Read 2 recent papers (2023–2025) published in IEEE RA-L on quadruped robotics and autonomous navigation topics. Noted structure differences from conference format — especially depth of literature review, experiment diversity, and discussion sections.	✓ **Completed**
Gap Analysis	Created a written gap list comparing the current conference paper against IEEE RA-L journal-level requirements. Identified 6 major areas needing expansion: literature review, experiments, figures, discussion, limitations, and abstract rewrite.	✓ **Completed**
Working Draft Setup	Created a Google Doc titled 'Journal Paper Draft'. Pasted full existing paper content inside and added section headings matching the IEEE RA-L template structure. This is the working document for the next 28 days.	✓ **Completed**

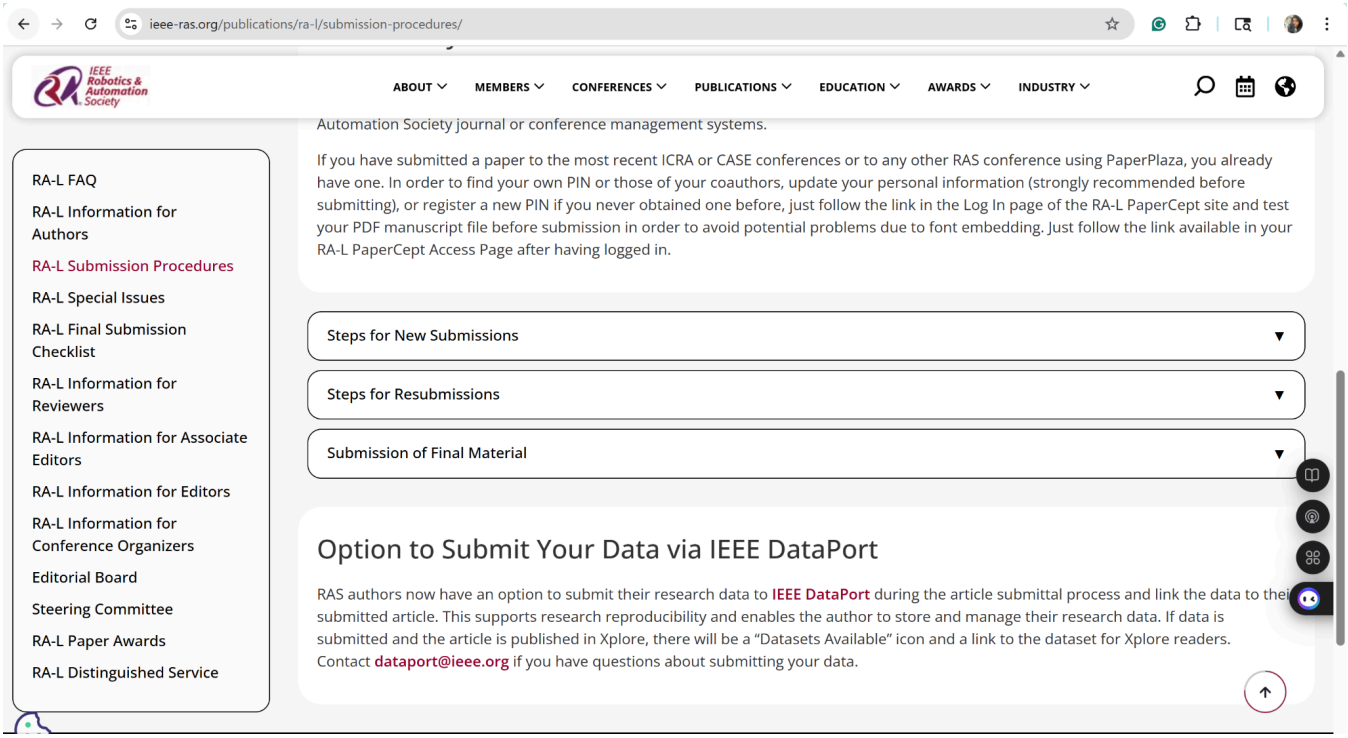
Screenshot Documentation	Screenshots of completed tasks were not fully captured during the session due to oversight.	✓ **Completed**

4. Gap Analysis Summary

GAP ANALYSIS: My Paper vs. IEEE Access Papers

1. **Literature Review Depth:** Paper 1 (Liu et al.) has 22 citations just in the introduction. I need to expand my "Related Work" section to include at least 15 more recent sources.
2. **Experimental Complexity:** Paper 2 (Murugan et al.) tests on "Unstructured Planetary Terrain." I need to move beyond simple flat-ground tests and document my robot's performance on uneven or cluttered surfaces.
3. **Hardware Specification:** Both papers include a detailed "System Architecture" with specific hardware labels (like the Aliengo robot and Jetson TX2 in Figure 3 of Paper 1). I need to add a labeled diagram of my 4-legged robot.
4. **Ablation/Comparative Study:** The journal papers compare their "Proposed Method" against "Existing Methods." I need to add a table comparing my navigation speed or success rate against a standard algorithm (like A* or basic ROS MoveBase).

5. Screenshots



IEEE Access

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APPLIED RESEARCH

Efficient Legged Robot Locomotion Through Optimized Gait Planning for Unstructured Planetary Terrain

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ABSTRACT Traditional rovers cannot maneuver easily through challenging environments, which limits their movement and exploration in planetary missions. Legged robots are potential solutions to replace them, but a detailed investigation is required to evaluate the mobility and the system efficiency. Control and gait-generation issues will have to be tackled to enhance the locomotion of a legged robot in such adverse environments. The paper will discuss the design and development of a Quadruped legged robot with emphasis on solving the most critical problems with mobility, control, gait generation, and power systems. The paper points to the possibilities of multifunction multi-modal mobile quadruped robots to be used in the exploration of planetary terrains, where their velocity and energy consumption are more advantageous than conventional rovers. Raising the state of the art in morphologies of leggings on different terrains and the use of sustainable and efficient on-board power generation systems are the key areas of development. The findings of the simulation suggest that the quadruped robot model was highly versatile as it was capable of climbing a 0.5-meter obstacle and traversing a 30-degree slope without losing its balance. The model showed that the gaits with full flight phases, like running trot/gait, become more efficient with reduced gravity. The findings confirm that the proposed system can afford high-speed and energy-efficient mobility to explore planets.

INDEX TERMS Adaptive gait planning, energy-efficient locomotion, gravity-adaptive control, legged locomotion, neural oscillators, planetary terrain navigation, quadrupedal robots, trajectory optimization.

I. INTRODUCTION

Autonomous systems development has formed the basis of traditional rovers as a go-to option for planetary missions [1]. Also, their maneuverability is limited by the uneven, unstructured surface of extraterrestrial environments [1], [2]. Alternatively, the agility and adaptability of legged robots compared to their biological counterparts make them an interesting option for traversing such difficult terrains [3], [4]. Specifically, quadruped robots can actively change their stance through the position of their feet, allowing them to pass obstacles and be stable on the surface where wheeled systems lose their grip [4], [5]. This versatility makes them suitable candidates for well-known complex missions in industrial inspection applications, disaster response situations, and planetary missions [6], [7], [8].

Despite their potential, the deployment of quadruped robots is hindered by the challenges in control, stability,

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RESEARCH ARTICLE

Safe and Robust Motion Planning for Autonomous Navigation of Quadruped Robots in Cluttered Environments

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ABSTRACT Quadruped robots, with their superior terrain adaptability and flexible movement capabilities, demonstrate greater application potential in complex environments compared to traditional ground robots. However, their non-negligible body shape and anisotropic motion characteristics complicate the achievement of high-precision motion planning and autonomous navigation. In this paper, we propose a safe and robust motion planning system tailored for autonomous navigation of quadruped robots in cluttered environments. We adopt a hierarchical architecture and decompose the planning process into front-end searching and back-end optimization. In the front-end searching stage, the robot finds a smooth, feasible, and energy-efficient initial trajectory with safety consideration. In the back-end optimization stage, we leverage B-splines to enhance the trajectory smoothness, safety, and motion stability. Finally, the time allocation is fine-tuned through iterative refinement, ensuring the feasibility of the optimized trajectory. Our method is extensively validated in challenging simulations as well as in real-world testing environments, benchmark comparisons also demonstrate the improved performance of our method.

INDEX TERMS Quadruped robots, autonomous navigation, motion and path planning, B-spline trajectory optimization.

I. INTRODUCTION

In recent years, owing to their superior flexibility, terrain adaptability, and payload capacity in unstructured environments, quadruped robots have gradually become a research focus in the field of robotics. They have achieved commendable outcomes in various applications such as terrain exploration, post-disaster search and rescue, material transportation, and industrial inspection, significantly reducing human labor costs [1].

Compared to traditional wheeled and tracked ground robots, quadruped robots possess significant advantages. On the one hand, they can adjust their gaits in response to different ground structures, enabling stable locomotion similar to quadruped animals in unstructured environments such as rugged terrains. On the other hand, they have approximate omnidirectional mobility to navigate through narrow spaces such as tunnels and caves by adjusting their body postures, thus exhibiting higher flexibility. Therefore, it is meaningful and promising to integrate autonomous navigation technology with quadruped robots to enhance their autonomy and further expand their practical application range.

Existing work on quadruped robots primarily focuses on novel mechanical structures [2], [3] and the study of their gaits motion, including gait generation, foot trajectory planning, as well as body balance and control robustness [4], [5], [6]. These studies ensure stable and efficient low-level motion for quadruped robots across different environments and tasks but they lack the research on their high-level center-of-mass motion behaviors, limiting their autonomous capabilities and range of application in task-oriented missions within complex environments. Moreover, the non-negligible body shape of quadruped robots, along with their movement capability difference in various orientations, introduces additional complexity and challenges to the motion planning and autonomous navigation.

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FIGURE 1. Composite image of an Aliengo robot navigating through the test field cluttered with boxes and boards.

FIGURE 2. The Aliengo robot (weight: 21.5kg, standing dimensions: 0.8m × 0.35m × 0.6m). It is equipped with a Slamtec mapper LiDAR for environmental perception, an NVIDIA Jetson TX2 for motion planning, and an onboard miniPC for motion control.

In this paper, we propose a novel motion planning framework for the autonomous navigation of quadruped robots in cluttered environments. Figure 1 shows a composite image of an Aliengo robot navigating through a cluttered test field with our proposed method. Our framework employs a hierarchical planning structure dividing the planning process into front-end and back-end stages. In the front-end stage, we propose a kinodynamic path searching method, named *Improved Kinodynamic A**, that integrates the robot's kinematics into the search process. This method balances the control energy consumption and the path smoothness to quickly search for a guiding global trajectory connecting the initial and terminal states. In the back-end stage, the guiding trajectory is parameterized into a B-spline trajectory, and we optimize the control points of the trajectory by leveraging the local controllability and convex hull properties of B-splines, thus enhancing its smoothness and improving clearance with obstacles. Finally, to ensure the feasibility of the generated trajectory, iterative refinement is employed, with the motion characteristics of quadruped robots taking into account, thereby eliminating infeasible velocities and accelerations within the trajectory.

We compare our method with an existing state-of-the-art work employing a similar planning method in both simulated and real-world environments. Benchmark comparisons demonstrate that our method has lower energy consumption and higher stability, with higher success rates in the autonomous navigation tasks. The main contributions of this paper are summarized as follows:

1) We propose a novel path searching method *Improved Kinodynamic A**, which generates a guiding initial trajectory conforming to the motion characteristics of quadruped robots.
2) Incorporating the body shape and motion characteristics of quadruped robots, we propose a trajectory optimization method based on B-splines to generate smooth, safe, and feasible trajectories.
3) Based on the optimized trajectory and the motion characteristics of quadruped robots, we propose an iterative refinement method that ensures the feasibility of the final trajectory.

The remainder of this article is organized as follows. Section II summarizes the related works. Section III introduces the quadruped robot hardware platform and the system architecture. Section IV details the *Improved Kinodynamic A** path searching algorithm, while Section V presents the gradient-based trajectory optimization algorithm. Section VI validates and analyzes the proposed method through both simulation and real-world experiments. Finally, Section VII concludes the article.

II. RELATED WORKS

As stated before, only a limited number of works have explored autonomous navigation of quadruped robots, including navigation in unstructured terrains or unknown environments [7], [8], [9], [10], [11], [12], [13], [14], the navigation with multi-gait combinations (e.g. walking and jumping) [15], [16], [17], and multi-robot cooperative exploration [18], [19]. For the motion planning and autonomous navigation of robotic systems, the prevalent approach is to utilize a two-stage hierarchical architecture, which can be decomposed into front-end searching and back-end optimization.

A. FRONT-END SEARCHING

Front-end searching, also known as global planning in some works, aims to search for a guiding global path in a known or partially known environment. The majority of existing works utilize search-based methods such as A* [7], [8], [9], [10], [15], D* Lite [13], or sampling-based methods such as RRT* [14] and RRT-Connect [16], [17] for global searching. However, while these approaches can find a guiding path geometrically, they lack the robot's kinematics, which may lead to infeasible trajectories for the actual robot, or increased energy consumption in the final trajectory [20]. To address this shortcoming, an intuitive solution is employing the kinodynamic searching. For instance, Hybrid-state A* [21] generates dynamically feasible trajectories for autonomous vehicles in the searching stage. Inspired by this, Zhou et al. [22] designed a kinodynamic path searching method for quadrotors. Considering the approximate omnidirectional mobility of quadruped robots within their motion plane, we propose the *Improved Kinodynamic A** approach,